

SUBHRONIL MUKHOPADHYAY

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EDUCATION

Vellore Institute of Technology | Vellore

September 2022 – May 2026

B.Tech. Computer Science (Final Semester) | CGPA: 8.88/10

SKILLS

Languages: Python, C++, C, Java, HTML, CSS, JavaScript, Motoko, R, MATLAB

AI/ML Tools & Libraries: TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, LLMs, RAG

Databases & Cloud: AWS, NLP, PostgreSQL, MySQL, MongoDB

Frameworks & Web Dev: OpenCV, ReactJS, NodeJS, jQuery, Bootstrap

Developer Tools: GIT, Figma, OrCAD, Verilog HDL, Data Structures & Algorithms (DSA)

Soft Skills: Networking, Leadership, Creativity, Research Skills, Strategic Thinking, Resilience

WORK EXPERIENCE

Hitachi India Pvt. Ltd., Bengaluru | Sr. Technical Engineer (Contract via Hitachi Systems) July 2026 – Present

- Engineered an agentic travel-planning middleware that turns a single natural-language prompt into a complete end-to-end itinerary, autonomously orchestrating train, hotel, and event bookings across 3 Model Context Protocol (MCP) servers exposing 6 tools.
- Architected a stateful 6-node LangGraph workflow with both deterministic and LLM-driven routing across 7 use cases, backed by a dynamic registry that discovers and spawns each MCP server as an isolated subprocess and reflects tool schemas at runtime.
- Produced a 1,000+ line Manim explainer video for GSK equipment-maintenance SOPs, animating a 5-stage process-discovery pipeline from raw demonstration capture through 3D embedding, normalization, and finite-state-machine (FSM) parsing.
- Engineered 3D scene rendering (ThreeDAxes, Dot3D, animated camera orbits) and an FSM resolving 10 canonical steps into 4 validated task orderings, translating a complex ML pipeline into a stakeholder-ready visual.

INTERNSHIPS

Hitachi India Pvt. Ltd., Bengaluru | Research and Development Intern

January 2026 – June 2026

- Analyzed 50K+ ATM error logs to identify recurring failure patterns; Constructed a predictive maintenance model that reduced unplanned downtime by an estimated 30%.
- Designed a rule-based verification system to flag missed or out-of-sequence steps in GSK equipment maintenance workflows, improving compliance tracking accuracy.
- Programmed a signal-processing module for real-time task validation using acoustic signatures, enabling touchless quality control on the factory floor.
- Architected a monitoring dashboard consolidating 10+ data center KPIs into a single operational view for the operations team.

BeyonData Solutions Pvt. Ltd., Ahmedabad | Machine Learning Intern

June 2024 – July 2024

- Evaluated 5 CNN architectures (VGG16, ResNet50, InceptionV3, MobileNetV2, EfficientNetB0) on a 27-class logo dataset across 7 metrics including F1-score, precision, and inference time.
- Engineered augmentation pipelines achieving 93.8% classification accuracy via MobileNetV2; synthesized results into a comparative framework to inform production model selection.

- Compared models across accuracy vs. inference time trade-offs to identify the optimal architecture for deployment.

PROJECTS

TimeCapsule Connect | MERN Stack, PostgreSQL July 2025

- Architected a full-stack platform supporting 100K+ time/location-based unlock combinations over 20+ years using KSUIDs and Google Maps API.
- Implemented a real-time collaborative editor (SlateJS + Yjs) with <100ms sync latency for 10+ concurrent users.
- Optimized 1 GB+ media uploads via chunked logic, cutting upload failure rates by 60%.

Dynamic Crowd Simulation | GNN, Transformer, PyTorch, PPO, OpenCV, Python March 2025

- Trained a hybrid GNN + Transformer model for pedestrian trajectory prediction, achieving 99%+ accuracy on the test set.
- Integrated a PPO reinforcement-learning agent for collision-free navigation; optimized custom reward functions to a mean score of +6.31 over 20K+ training timesteps.

Advanced Voting System | HTML, CSS, JavaScript, NodeJS, OpenCV, TensorFlow, Keras, Python May 2024

- Engineered a secure voting platform with Digilocker and Voter ID authentication, targeting 1,000+ remote users and a projected 20% increase in voter turnout.
- Stress-tested for 10K+ daily visitors, validating stable performance under peak load conditions.

Player Injury Prediction | ANN, TensorFlow, Keras, Python August 2023

- Trained an ANN on NBA performance data reaching 89% accuracy and 90% precision for injury-type forecasting.
- Identified injury trend patterns contributing to a projected 25% improvement in prevention protocols and 40% reduction in at-risk player downtime.

CERTIFICATIONS

GATE Qualified – GATE, Mar 2026

Amazon ML Summer School – Amazon, September 2025

Microsoft Azure AI-900 – Microsoft, August 2024